

AquaSync

When should a dam let water go? *Before* the storm, not after it.

A flood-management twin for any basin where reservoirs, rivers and the tide interact. It simulates the connected water bodies together, searches release policies on a forecast, and hands a named operator one sentence to approve or reject. Its first proving ground is Kerala's Periyar basin, the case study throughout this document.



Photos and videos of the project

drive.google.com/drive/folders/18p7ca7HLUWT4xghb60SKdXnopAxLefih

OPEN →

REPLAY ERROR · CASE STUDY

0.30 m

Mean level error over 20 days of October 2021, against the KSEB bulletin.

FLOOD CUSHION

about 3 m

More than the day had, in hindsight, with slightly more revenue. About half a metre of it sits inside model error.

TESTS

139

Physics and behaviour tests, collected today. One command, `check.py`, is the gate.

ANY DAM-RIVER-SEA BASIN · CASE STUDY: PERIYAR, KERALA

A NAMED OPERATOR APPROVES EVERY RELEASE

ADVISORY, PERMANENTLY

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Periyar case study: six numbers

0.30 m

Replay error over 20 days of October 2021; 0.57 m at worst.

0.32 m

On August 2022, an episode never tuned to. The drift changes sign.

about 3 m

More cushion than the day at every lead time. About 0.5 m sits inside model error.

61% → 40%

Share of released water spilled, at zero lead versus 30 days.

48 h / 90 h

Forecast matches hindsight to here (2021 / 2022), then +69% or +158% excess cost.

911

cumecs at the confluence with each dam optimised alone, against 403 observed. Two dams' contribution only.

SEE IT BUILT

Photos and videos

Photographs and video recordings of the project are kept together in one shared folder. Open it alongside this document: the pages here explain what the pictures show.

[Open the Google Drive folder →](#)

<https://drive.google.com/drive/folders/18p7ca7HLUWT4xghb60SKdXnopAxLefih?usp=sharing>

The project in one breath

Most reservoirs are run on rule curves that react to a level once it is crossed, one dam at a time. Floods do not respect that: rain, reservoirs, rivers and the tide act as one connected system. In October 2021 on the Periyar, Idukki entered a storm above its own rule level and both dams opened on the same day. AquaSync replays such episodes in a physics-based twin, then asks what a release schedule chosen on a forecast would have done instead. It never touches a real gate: the output is advice, for a named person to accept or refuse.

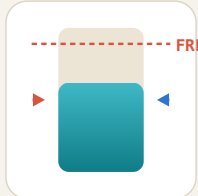
Full for power, empty for floods

Every reservoir above a river faces the same conflict. The rain triggers a flood; the timing of the releases decides how much of it becomes a disaster.

DISASTER MANAGEMENT

wants it LOW

Empty storage is the only flood cushion that exists.



THE POWER UTILITY

wants it FULL

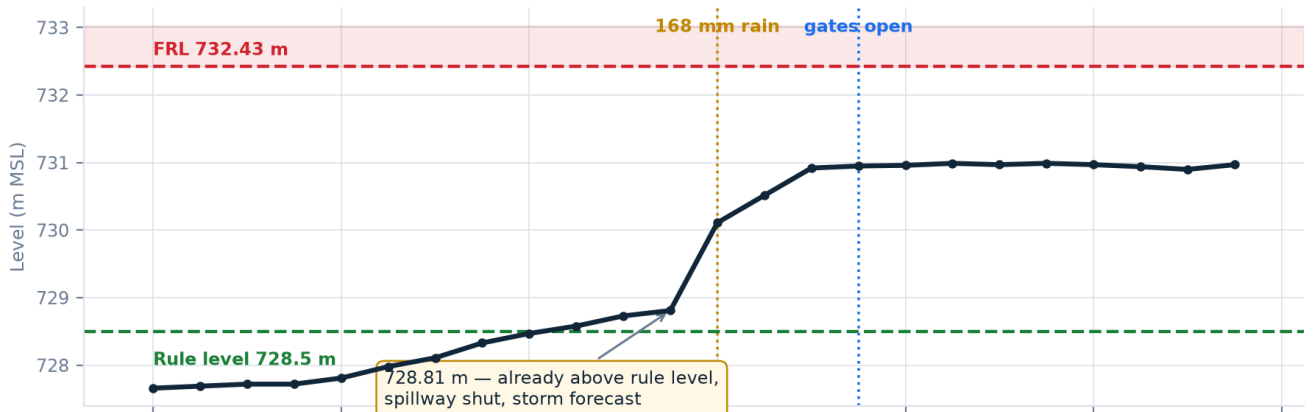
Head is revenue. Water released without generating is money poured away.

- 1 Release too early** and the dry season is short of water that was needed.
- 2 Release too late** and every gate opens at once, into a river that is already full.
- 3 The tide decides** whether a release reaches the sea. At high tide the Arabian Sea holds the lower Periyar up.

The gap. No operational system in Kerala today weighs these together, on a forecast, across more than one reservoir at a time.

Case study: the Periyar, October 2021, in public data

Idukki reservoir level, October 2021



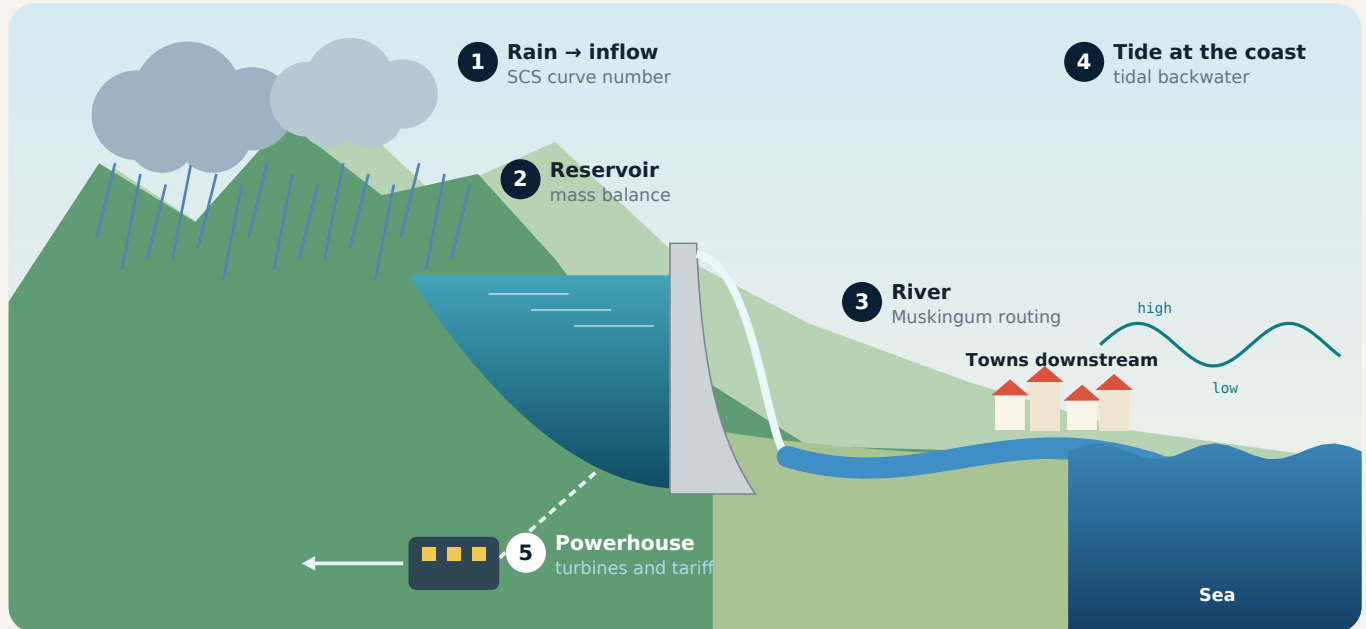
Inflow and spillway release



Idukki, October 2021. The reservoir met the storm of 16 October at 728.81 m, already above its own 728.5 m rule level, with the spillway shut. Inflow peaked at about 879 cumecs. The level reached 730.99 m, 1.4 m from full reservoir level, and the spillways of both Periyar dams opened on 20 October, into the same river on the same day. Nothing was concealed and no rule was broken: the level rose faster than a reactive procedure can respond. Source: KSEB daily bulletin.

Every water body in the chain, together

Five physical models run in sequence, hour by hour, for whatever basin they are given. The chips say how far each is validated on the Periyar case study.



PER BASIN storage curve from bulletins · catchment from a DEM · land-use curve number · reach geometry · nearest tide port

- 1 Rainfall → inflow**
SCS curve number with a triangular unit hydrograph
 $Q = (P - 0.2S)^2 / (P + 0.8S)$

Antecedent wetness dominates: the curve number climbs as a long spell saturates its own catchment, which is what turns a wet week into a flood.

volume -1% pooled, -17 to +38% by season · NSE 0.07
- 2 Reservoir mass balance**
Hourly Euler, fitted level-storage power law
$$\frac{dV}{dt} = Q_{in} - Q_{turbine} - Q_{spill} - Q_{evap}$$
$$S(h) = S_{FRL} \cdot ((h - h_{dead}) / (h_{FRL} - h_{dead}))^\beta$$

β above 1 means storage grows faster than height, so the top metres of a reservoir are worth the most as cushion.

β 1.348 · r^2 0.996 · MAE 17 Mm³ on 1,836 rows
- 3 River routing**
Muskingum, split into stable sub-reaches automatically
 $2Kx \leq \Delta t \leq 2K(1 - x)$

A long reach at an hourly step breaks this bound and the river appears to run backwards; splitting it into sub-reaches is the correct discretisation.

anchored to CWC's 8 h travel time · not gauge-calibrated
- 4 Tidal backwater**
Harmonic tide; Kochi M2 S2 K1 O1 N2 here
 $effective\ conveyance = f(Q, tide)$

About a metre of spring range decides whether a full river overtops, and it recurs twice a day: a free window in which the same volume moves at lower risk.

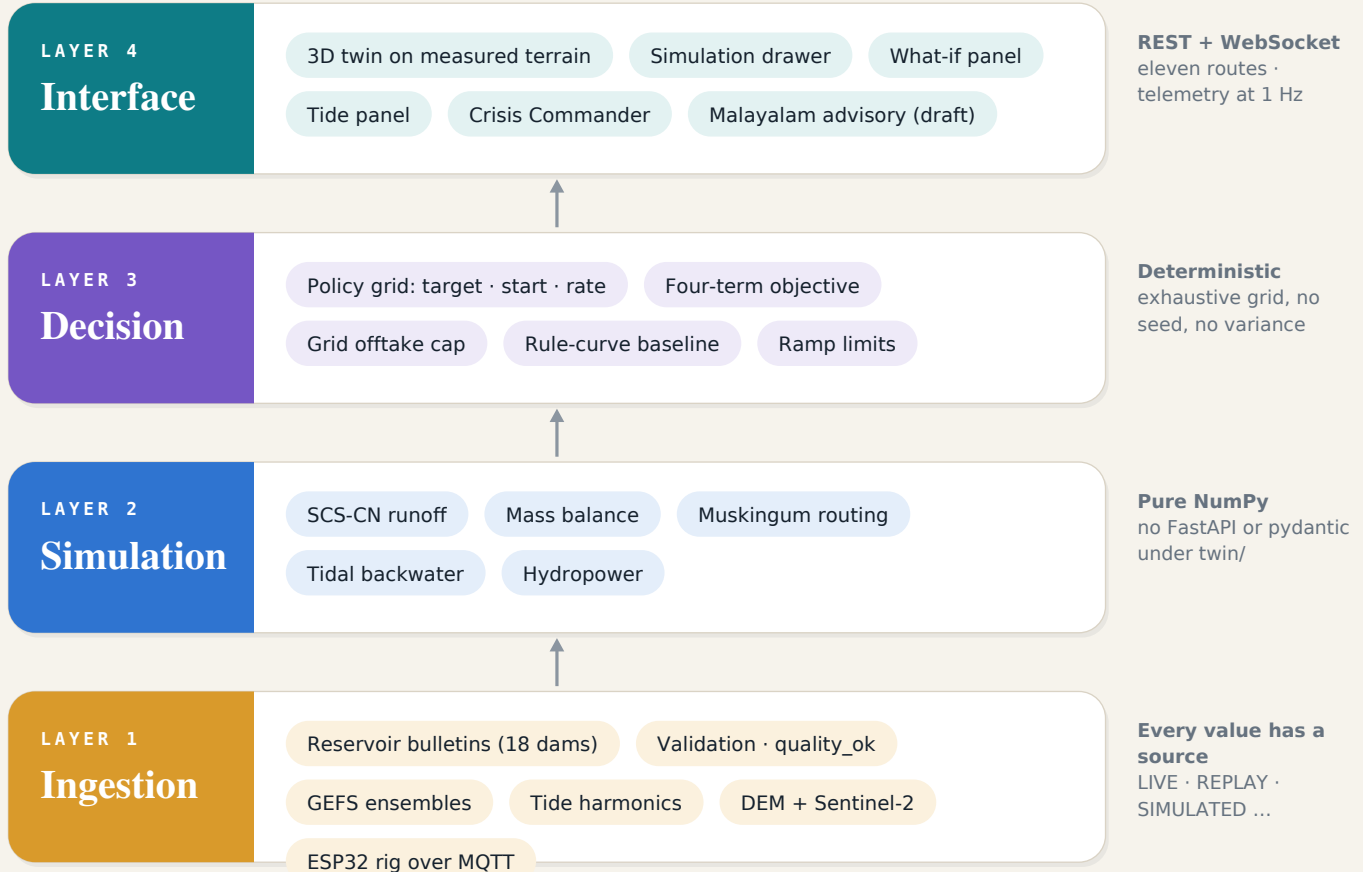
sanity-checked against published descriptions · not validated
- 5 Hydropower and tariff**
Hill-diagram efficiency, time-of-day tariff
 $P = \rho g Q H \eta$

Only spill the turbines could have absorbed counts as lost generation. At Idukki the powerhouse discharges into another river, so generation never loads the Periyar.

indicative KSEB time-of-day bands, not a current tariff order

Four layers and one boundary

The simulation core is pure NumPy and imports no web framework; continuous integration enforces it. That is what makes the physics testable on its own.



The trust boundary



Liability

If an automated system opens gates and someone drowns, there is no acceptable answer to who is accountable. A recommendation a named officer approves has one.

Trust is earned in shadow mode

The credible path is a full monsoon running alongside practice, logging what it would have advised against what happened.

The model is wrong in known ways

Daily inputs, uncalibrated routing, no 2D inundation. Acceptable in an adviser; unacceptable in an actuator.

PROVENANCE ON EVERY READING



Only LIVE is green. A rig reading is SCALE_RIG and never converted to metres above sea level: a 400 mm tank is not a reservoir.

Search policies, not schedules

An operator cannot execute eight hundred hourly setpoints. They can execute a sentence. So the search runs over the three numbers a control room actually uses.



THE OBJECTIVE

$$\begin{aligned}
 J = & w_{\text{flood}} \cdot \Sigma(\text{overtopping})^2 \\
 & + w_{\text{safety}} \cdot \Sigma(\text{FRL encroachment})^2 \\
 & + w_{\text{revenue}} \cdot (\text{forgone generation}) \\
 & + w_{\text{gate}} \cdot (\text{ramping} + \text{gate movements})
 \end{aligned}$$

Flood cost is squared because damage grows faster than depth.

Weights are policy, not constants

Choosing them is a public decision. They are exposed so an operator can see and set them rather than inherit a value judgement buried in code.

The grid decides what can be sold

Without the offtake cap the optimiser runs the turbines flat out for a month and books revenue the grid would never take.

A sudden release is itself a hazard

People and livestock are in the riverbed. A schedule that ramps faster than staged opening allows is not implementable.

Why not hourly setpoints?

The first version searched them. Ten days looked better than fourteen, which looked better than twenty-one: the search covered longer horizons more sparsely. It measured luck, not lead time.



WHAT THE OPTIMISER HANDS OVER • OCTOBER 2021, IN HINDSIGHT

Draw Idukki down to 727.96 m, starting at hour 207 of the episode, releasing no more than 828 cumecs.

Hindsight means it saw the inflow that actually arrived. With a real forecast, see §7.

Replayed against the record

Before the twin is allowed to recommend anything for a basin, it has to reproduce what actually happened there. On the Periyar case study, it did, on data nobody tuned it to.

OCTOBER 2021 REPLAY

0.30 m

Mean level error over 20 days; 0.57 m at worst.

AUGUST 2022, OUT OF SAMPLE

0.32 m

An episode never tuned to. The drift changes sign between the two storms.

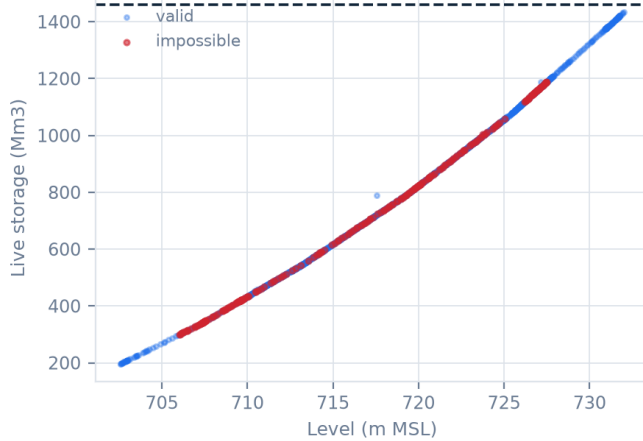
LEVEL-STORAGE FIT

r^2 0.996

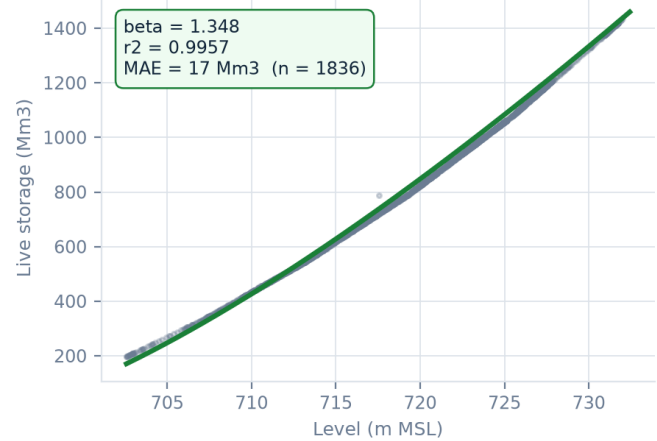
β 1.348, MAE 17 Mm³ on 1,836 validated bulletin rows.

Raw feed — 11% of rows are impossible

physical capacity at FRL



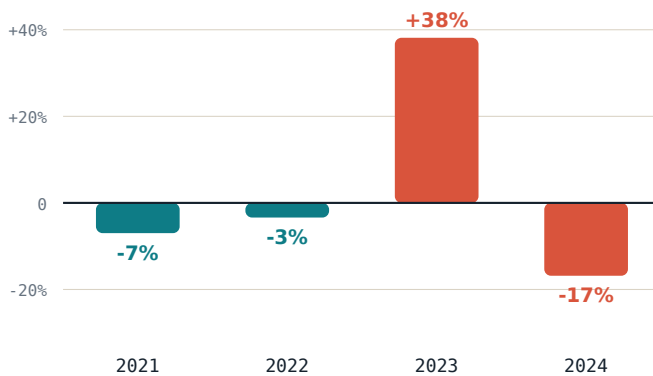
Validated rows — fitted power law



Level against storage from the KSEB bulletin, fitted as a power law. The validation layer sets aside physically impossible rows first; about 11% of the feed fails, in one contiguous block caused by a column-alignment bug upstream.

RAINFALL-RUNOFF, FOUR MONSOONS

Volume error by season



Pooled -1%, but the seasons swing. r^2 0.55 on shape, NSE 0.07 on amplitude: no recession limb. Calibration fails leave-one-season-out, so handbook CN 72 stays.

CATCHMENT FROM A DEM

570 km²

net of the Mullaperiyar diversion, against the 650 km² in use (-12%). Main channel about 66 km. A DEM sees the hills, not the tunnel.

ROUTING CALIBRATION · BLOCKED

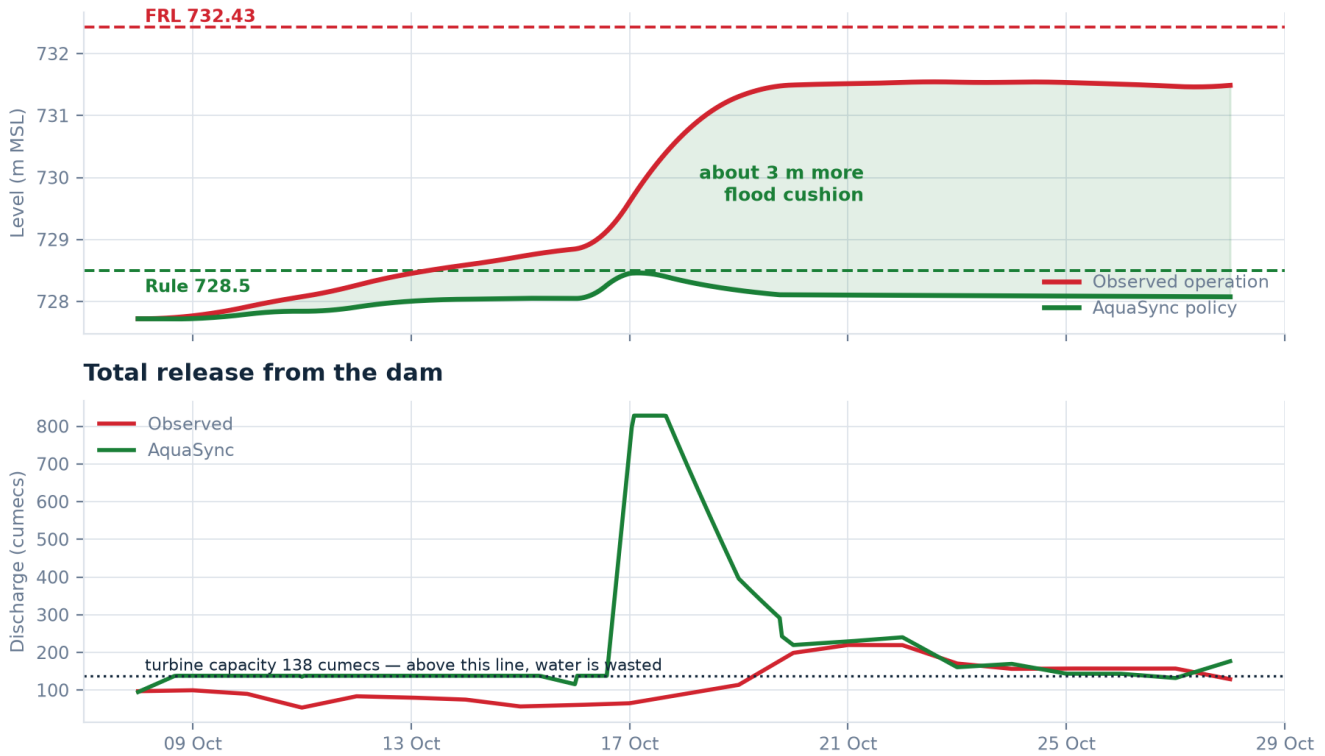
r^2 0.005

on 1,967 days of CWC gauge data. A daily step is about three times coarser than an 8 h travel time, and releases are daily-only. The CWC anchor stands.

About three metres of cushion

Replayed against the schedule the optimiser would have chosen, October 2021 ends with about 3 m more room below full reservoir level, and slightly more revenue, not less.

Reservoir level: what happened vs what the twin recommends

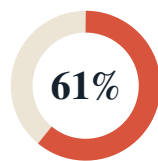


CUSHION, AT EVERY LEAD TIME

about 3 m

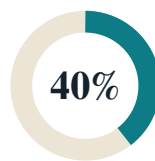
The range across lead times is 2.7 to 3.3 m, and roughly half a metre of it sits inside the twin's replay error, so it is quoted in whole metres.

RELEASED WATER THAT IS SPILLED



zero lead

act on the day



30 days

act a month ahead

Lead time changes the waste, not the cushion.

REVENUE, ENERGY-NEUTRAL

+₹4.4–9.6 cr

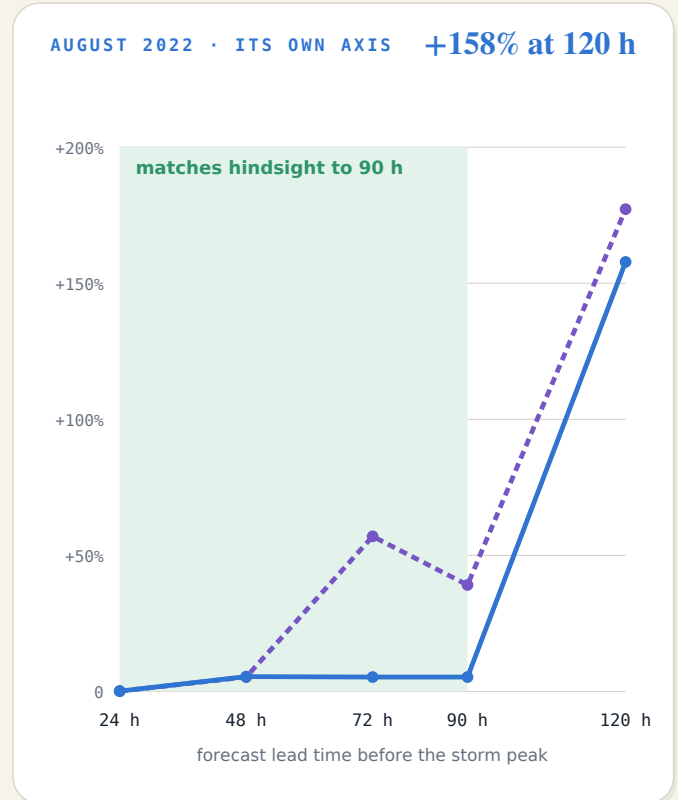
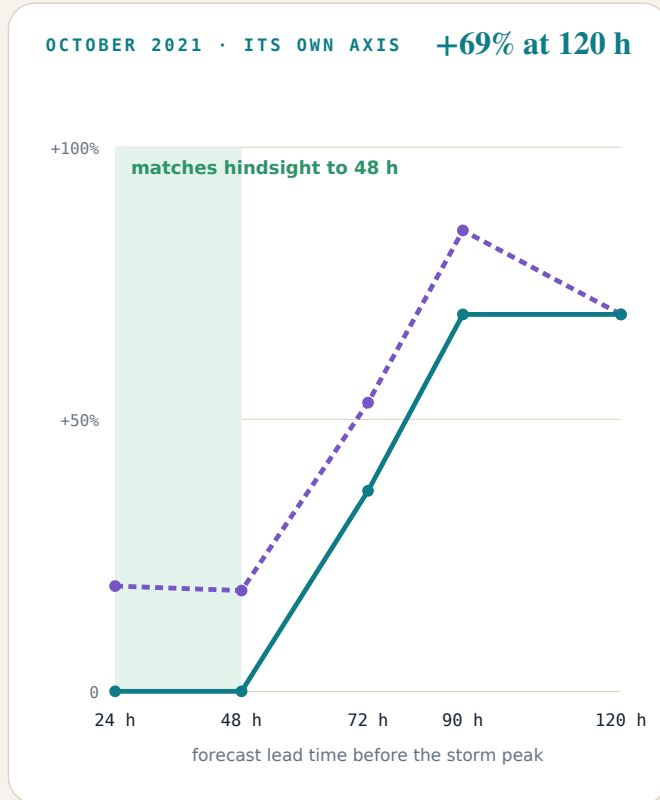
The same water, shifted into higher-tariff hours. Indicative KSEB time-of-day bands, not a current tariff order; horizons differ, so rows are not strictly comparable.

READ WITH IT

It releases more, earlier. The optimised schedule's routed peak below the dam is about 828 cumecs against 220 for the day: one dam, an uncalibrated reach, never read against bankfull. **It is hindsight.** It saw the inflow that arrived. **It does not always win.** On August 2022, a storm with no flood risk, the hindsight schedule holds about 1 m less cushion than the day: with nothing to protect, the objective prefers holding water, and the day's operators did fine.

How far ahead can it be trusted?

Hindsight is free. The honest test is to hand the optimiser the rainfall forecast that was actually issued, then score what it chose against what perfect foresight would have.



— expected-value policy - - - hedging (minimax regret)

y axis: excess cost against perfect foresight, on the optimiser's own objective

A ramp, then a cliff

October 2021 degrades gradually from 48 h. August 2022 holds to 90 h, then jumps to +158%. Two storms, two shapes, so they are never averaged into one curve.

Hedging rarely pays

The minimax-regret policy beat the expected-value one in 1 run of 10, by 0.16 points. In every other run it cost the same or more.

The right yardstick

Excess cost is zero when the forecast picked the hindsight policy and is never negative. Freeboard retention is not quoted: it can pass 100% while giving revenue away.

WHAT IT MEANS FOR A CONTROL ROOM

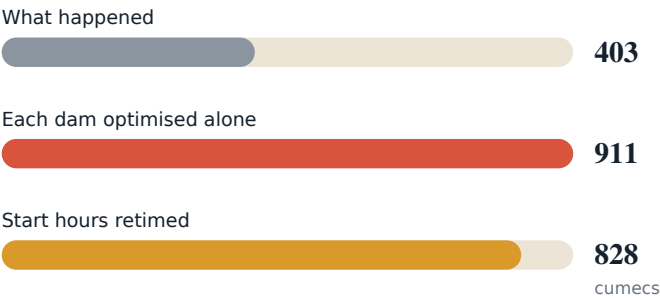
On storms like these, a release decision taken on a 48-hour forecast is about as good as one taken with perfect knowledge of the rain. Beyond that the answer depends on the storm, which is exactly why the recommendation stays advice.

What did not work, and what it taught

Four claims on this project were stated early and broken by the next data point. The corrections are kept on the record, and these are the results that stayed awkward.

CASCADE · TWO DAMS, ONE RIVER

Optimising each dam alone is worse



Joint peak at the confluence, October 2021. Alone, the dams put it 126% above what happened; retiming both recovers 9%. **Two dams' contribution only, never read against bankfull.** It is an objective-function problem.

THE FIX, BUILT · AND INERT

Joint cascade objective

One shared flood term on the combined discharge at the confluence; each dam keeps its own safety, revenue and gate-wear terms.

0 moves

On the data as held the shared term is identically zero, so coordinate descent converges without moving. The blocker is the ungauged lateral inflow. An *assumed* lateral (not a measurement) first changes the policies at 300 cumecs.

STORM STRESS TEST · A SENSITIVITY, NOT A RETURN PERIOD

Multiply the recorded storm. When does each schedule reach full reservoir level?

EPISODE	THE DAY'S SCHEDULE	AQUASYNC'S
Periyar cascade, October 2021	at ×1.25	not by ×3
Idukki riding FRL, Nov-Dec 2021	at ×1.25	not by ×3
Idukki monsoon spill, August 2022	at ×1.75	at ×3

A scaled copy of one hydrograph: same shape, more water. The optimiser may release up to 1,500 cumecs, so holding the reservoir moves water downstream instead. The ladder is coarse, so each multiple is an upper bound.

! **A runoff chain that produced no runoff**
The curve number's initial abstraction was charged every timestep, so the answer depended on the step. Found on a task that looked small; fixed, and pinned by a test.

! **A LIVE badge on a 2021 recording**
The socket being open said the backend was reachable, not that anything was measured. The server now declares provenance on every frame.

! **Routing calibration, r^2 0.005**
Daily release data cannot resolve an eight-hour travel time. Blocked until the CWC 15-minute feed exists; the published anchor stands.

! **A BOM whose total disagreed with its rows**
Corrected, and this document now sums the rows itself and refuses to build if they disagree.

A twin you can stand in front of

The dashboard runs from a laptop with the network cable pulled. The ground is a satellite photograph laid on real terrain; the water rises and falls with the replay.



- 1 Reservoir level against dead, rule and full levels
- 2 Site, gate and basin views
- 3 A caption saying what is measured and what is drawn
- 4 Advice in English and Malayalam
- 5 Source: replay, recorded episode, not live



The simulation drawer scrubs the episode hour by hour: recorded level, the twin's replay and the optimiser's schedule on real axes, driving the 3D water.

What-if

Drag a release and the card answers from the API: 72 h peak level, cushion, and advice.

Crisis Commander

You get the briefing the operators had and give the order. The same optimiser scores you, the day and AquaSync side by side.

Tide panel

The 72 h harmonic tide at Kochi with low-water windows shaded, labelled PREDICTED.

Catchment policy

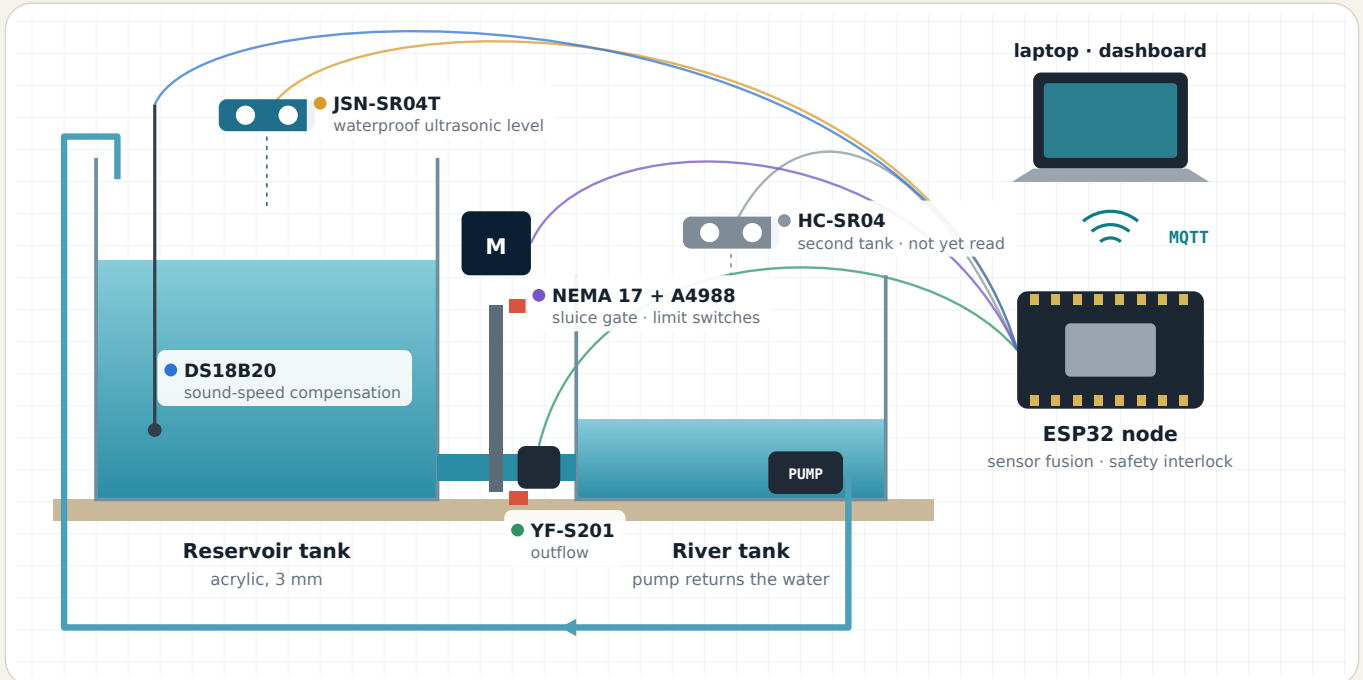
Change the curve number and see runoff volume. The peak is greyed: shape is not validated.

Malayalam advisory

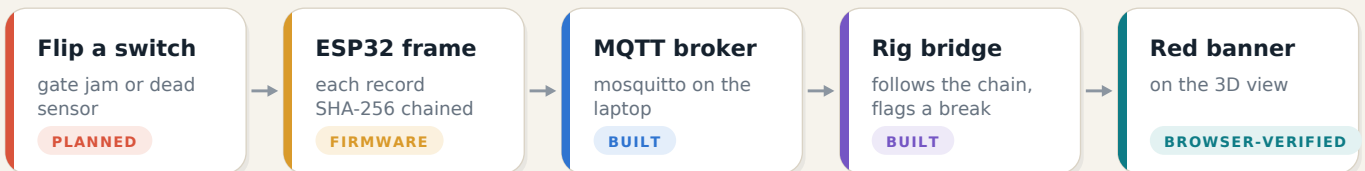
A line on every frame, on the same KSDMA bands. Draft wording: a native speaker must read it before public use.

A thirty-centimetre dam on a bench

The rig closes the loop onto a motor because it is an acrylic tank, not a reservoir. Components are in hand; the bench is not built yet and the firmware is untested on hardware.



The beat that wins the room: fault injection



ACTUATOR DISAGREEMENT – commanded 70%, verified 21%

Shown in a real browser against a simulated node publishing hash-chained frames over mosquitto. The reservoir stayed labelled REPLAY while the rig read LIVE, so the two provenances never blur. What is left is the physical switches and the firmware paths behind them.

When the network dies



433 MHz is the legal ISM band in India; the 868 MHz modules widely recommended online are the European allocation.

The V1 rig, for ₹6,250

Every part, from hardware/bom/README.md. Prices are indicative, sourced from Indian vendor listings in August 2026; budget $\pm 25\%$.

#	COMPONENT	SPECIFICATION	QTY	₹
1	ESP32-WROOM-32 DevKit	38-pin, dual-core 240 MHz, Wi-Fi + BT	1	450
2	JSN-SR04T v3.0	Waterproof ultrasonic, 25–450 cm, ± 1 cm	1	550
3	HC-SR04	Ultrasonic 2–400 cm	1	90
4	DS18B20 (waterproof probe)	1-Wire, $-55\ldots 125$ °C, ± 0.5 °C	1	180
5	BMP280	Barometric, 300–1100 hPa, ± 0.12 hPa	1	150
6	NEMA 17 stepper	42×42 mm, 1.8°, 4.2 kg·cm	1	650
7	A4988 driver + heatsink	1/16 microstep, 2 A	1	130
8	YF-S201	Hall flow sensor, 1–30 L/min, G1/2"	1	350
9	12 V submersible pump	800–1000 L/h	1	300
10	12 V 5 A adapter	Barrel jack	1	450
11	LM2596 buck	12 V→5 V, 3 A, adjustable	1	120
12	Micro limit switches	SPDT lever	2	80
13	Acrylic sheet, 3 mm	2 × (600×450 mm)	2	900
14	Acrylic solvent cement	Dichloromethane, 50 ml	1	200
15	Silicone tubing + fittings	8 mm ID, 2 m, elbows, clamps	1	350
16	Perfboard + headers	10×15 cm, assorted headers	1	200
17	Jumper wires	M-M / M-F / F-F, 120 pc	1	200
18	Breadboard	830 point	1	150
19	Micro-USB data cable	Data, not charge-only	2	200
20	IP65 enclosure	150×100×60 mm	1	350
21	Assorted R/C kit	1/4 W resistors, ceramics	1	200

21 line items · rows summed by this builder

₹6,250

ESP32 PIN MAP



V1 ₹6,250

Hardware-in-the-loop
scale rig

1 week · the core build

V2 +₹2,300

Off-grid resilience

+1 week · optional

V3 +₹5,900

Edge AI and better
sensing

+2 weeks · optional

V4 +₹9,900

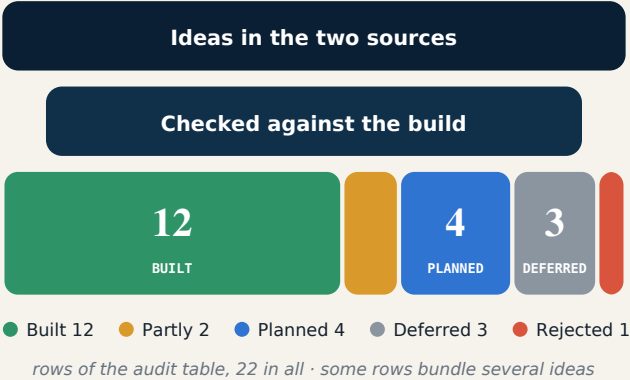
Offline command post
and bathymetry

+3 weeks · optional

Where the ideas came from

AquaSync began as one line in a planning conversation: a digital twin for the rain, the dam and the river, connected, that could find the best time and quantity of water to release.

Two source documents seeded the project and still sit in reference/source_chats/: a 17-page brief exported from an AI planning conversation, and a team chat export beside it. They proposed far more than any team could build in a season, from a river rover to post-quantum SCADA. The project's rule is that nothing in them goes unaccounted for. Each idea was checked against what is built on 11 September 2026, and each has a status and, where it was set aside, a recorded reason.



What the brief proposed, and what the project did with it

THE BRIEF SAID

AQUASYNC DID

Runoff as $C \cdot I \cdot A$, a single coefficient	→	SCS curve number that shifts with antecedent wetness, validated on four monsoons
Closed-loop gate control on the dam	→	Advisory, permanently. The only gate actuated is the bench rig's model sluice
Open gates before the rain, at low tide	→	Tide panel and tidal conveyance in the optimiser's flood term
Replay the 2018 crisis to prove it	→	October 2021, because the public dataset holds no 2018 rows. Correction on the record
LoRa / ESP-NOW disaster mesh	→	V2 tier: two 433 MHz SX1278 modules ordered; bench-only until V1 runs
Adaptive siltation curve, cavitation FFT, 2D inundation, river rover, edge LSTM	→	Deferred with reasons in ROADMAP.md, so they cannot creep into the build
Validate on the public Kerala dam dataset	→	About 11% of it is physically impossible; a validation layer flags every row
Tamper-evident release logging	→	SHA-256 record chain in the firmware, verified by the rig bridge

Every idea, accounted for

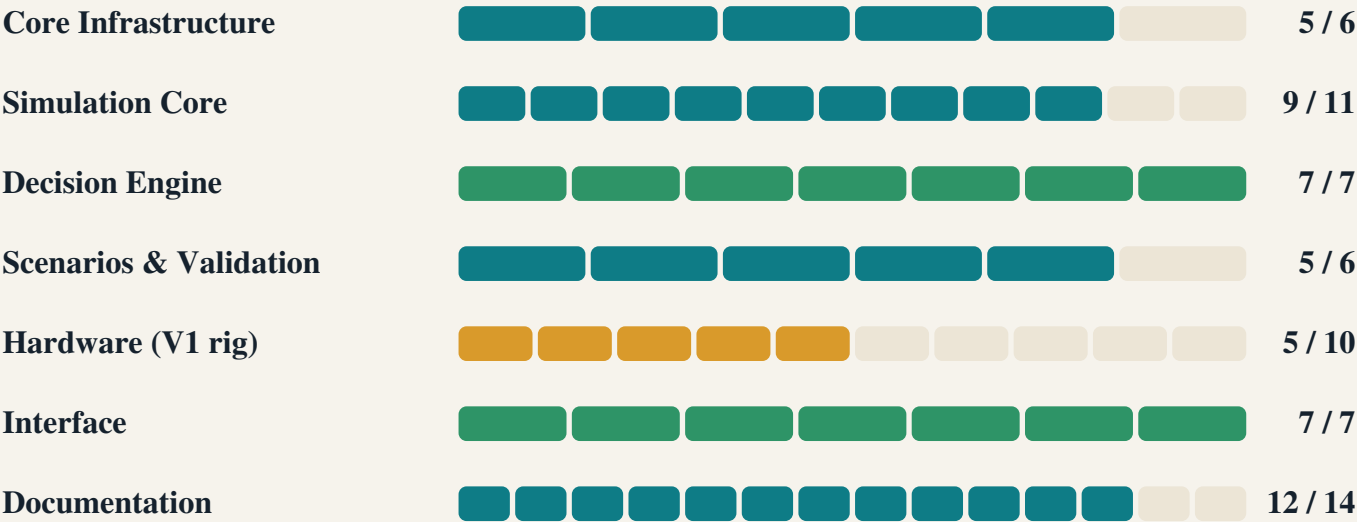
The reference-source audit from ROADMAP.md, 11 September 2026, reproduced row for row.

Mass-balance twin, SCS-CN runoff, Muskingum lag, tidal backwater, hill-diagram hydropower, multi-objective release policy, grid offtake	BUILT	Built and validated
Pre-emptive discharge timed to low-tide windows	BUILT	Built
"Replay the historical crisis, then run the optimiser" demo	BUILT	Built
2D/3D map, live gauges, recommendation panel, what-if slider	BUILT	Built
Timeline slider to scrub the episode, visitors try alternatives live	BUILT	Built 11 Sep
Colour-coded flood zones on a map	PARTLY	Reservoir bands only (dead / rule / FRL)
Cascade of dams as a graph with lags	BUILT	Built for the two dams that matter
Tamper-evident release ledger (hash chain)	BUILT	Built
Shadow-mode SCADA path, advisory only	BUILT	By design
Offline-first dashboard on a laptop hotspot	BUILT	Built and rehearsed
Crisis Commander serious game	BUILT	Built
EKF / sensor fusion on the node, physical fault-injection switches, predictive gate-jam detection (INA219 current spike)	PLANNED	Plan, hardware
LoRa fail-safe mesh	PLANNED	Plan, V2 pulled forward
Barometric squall pre-detection (BMP280 dP/dt)	PLANNED	Plan, hardware
Rain gauge on the upstream node	PLANNED	Plan, hardware
Synthetic 500-year storms, Monte Carlo probability of failure	PARTLY	Built as a sensitivity, not a probability
Land-use "policy mode" (curve-number slider)	BUILT	Built 11 Sep, volume only
Malayalam last-mile alerting	BUILT	Built 11 Sep as template wording
Economic loss calculator, citizen photo gauge, adaptive siltation curve, demand forecasting in the objective	DEFERRED	Deferred, with reasons
2D inundation, SAR calibration, evacuation routing and agent-based evacuation, GNN routing, RL gate policy, PINN surrogates, bathymetry boat, dam-breach mode, thermal seepage, hydrophone and cavitation acoustics, WebXR/AR, post-quantum SCADA, Malayalam social sentinel, TinyML sensor health, LSPIV and camera gauges	DEFERRED	Deferred, with reasons
LDMC and Kudumbashree mapping, OGC SensorThings output, delivery of the Malayalam line over a real channel	DEFERRED	Later
Insurance premiums, nudge dashboard, smartphone seismic network, irrigation-canal actuation, FPGA gate interlock	REJECTED	Rejected

Where a status reads 'deferred' or 'rejected', the reason is recorded in ROADMAP.md's deferred table. Deferral means later; the separate never-in-scope list means never.

50 of 61 components done

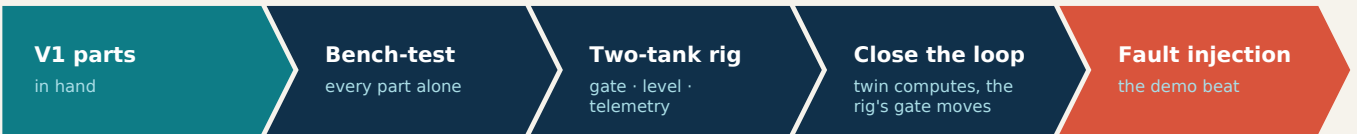
Counted from PROGRESS.md on main, last updated Friday 11 September 2026. The software is finished and validated; the bench rig is the critical path.



Phase map



The critical path



NEVER IN SCOPE

Operating a real dam gate. Kerala's gates are run by KSEB and the district administration under a statutory chain of accountability. There is no path from this system to them and there will not be one.

Presenting simulated values as live. Every reading carries its provenance.

DELIBERATELY DEFERRED

2D inundation, Sentinel-1 flood extent, evacuation routing, graph-network routing, reinforcement-learning gate policy, physics-informed surrogates, a bathymetry boat, dam-breach mode, acoustic cavitation sensing, AR, TinyML sensor health, camera gauges. Each is in ROADMAP.md with the reason it waits, so it cannot slip into the build.

Reproduce every number

Nothing in this document was typed in. It is rebuilt from the repository, and the gate fails if a rebuild changes it.

```
python scripts/check.py           # lint, tests, glyphs, regeneration, determinism
python scripts/build_showcase.py  # this document
cd backend && python -m pytest --collect-only -q
```

CLAIM	SCRIPT	DATA/PROCESSED
Level-storage fit	make_figures.py	figure_facts.json
October 2021 and August 2022 replays	out_of_sample_replay.py	replay_*.json
About 3 m of cushion	lead_time_study.py	lead_time_headline.json
Forecast error, two storms	forecast_error_study.py	forecast_error_study_*.json
Cascade and joint objective	cascade_*.py	cascade_*.json
Runoff validation	runoff_validation.py	runoff_validation_idukki.json
Routing calibration	routing_calibration.py	routing_calibration_*.json
Storm stress test	stress_sweep.py	stress_sweep_*.json

Data

KSEB / KSDMA daily reservoir bulletin · CWC Neeleeswaram gauge · NOAA GEFS forecast ensembles · IMD gridded rainfall · AWS Terrain Tiles · Copernicus Sentinel-2 (contains modified Copernicus Sentinel data) · Kochi tidal harmonics from published constituents.

Photos and videos of the project
<https://drive.google.com/drive/folders/18p7ca7HLUWT4xghb60SKdXnopAxLefih?usp=sharing>

AquaSync

Built by the AquaSync team
Status as of Friday 11 September 2026 · 139 tests

ADVISORY, PERMANENTLY. AQUASYNC NEVER OPERATES A REAL DAM GATE.